Machine Learning Basics

What Is Machine Learning

Machine learning is a branch of artificial intelligence in which computers learn patterns from data instead of following hand-written rules. A model is shown many examples, adjusts its internal parameters to reduce its errors, and then makes predictions on data it has not seen before.

Supervised Learning

In supervised learning every training example comes with a label. Regression predicts a number, such as a house price, while classification predicts a category, such as whether an email is spam. Common algorithms include linear regression, logistic regression, decision trees, random forests and gradient boosting.

Unsupervised Learning

Unsupervised learning works with data that has no labels. Clustering groups similar examples together, for example customer segments, and dimensionality reduction methods such as PCA compress many features into a few that keep most of the information.

Overfitting and Validation

A model that memorises its training data performs well in training but poorly on new data. This is called overfitting. Splitting data into training, validation and test sets, using cross-validation, and adding regularisation all help to detect and reduce it.